

Features and Distractors

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Prove that one measurement is insufficient for Features and useful distinctions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Features and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Training covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

All animal books in training have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

6. Solve the given case

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful evidence and possible distractions.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Predict whether a book is short using page count and cover colour. Which feature directly supports the defined label? Give a counterexample to colour.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 9 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use explainable traits that connect to the reader's stated preferences.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

All animal books in training have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Separate inputs from the target

For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality. Choose the measurements needed as inputs from length, width, colour, card number and the final inspection sticker. State the target label and write the feature record for a 12 cm by 8 cm red card using a defined order.

My reasoning and evidence

13. W2 Prove that one measurement is insufficient

Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

My reasoning and evidence

14. W3 Challenge an apparent colour shortcut

A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels. Explain why the perfect practice score did not establish that colour was a reliable feature for this task.

My reasoning and evidence

15. W4 Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name. A pass sticker is added afterward. Choose inputs with units. Construct two tiles with the same height but different labels. Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an input before checking.

My reasoning and evidence

16. W5 Notice when the task changes

The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded feature becomes useful? Explain why length and width alone may now be insufficient. Give two cards that support your explanation.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Features and Distractors

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Training contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test outputs have not been supplied.

AI ACADEMY / AI-2/9

Features and Distractors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Spot confounding

Topic and logo always move together in the training set.



2. Hold topic fixed

Compare the same mystery description with triangle versus circle logo.



3. Hold logo fixed

Compare mystery versus science descriptions with identical triangle logo.



4. Observe separately

Record predictions before concluding which clue the classifier uses.

CHANGE ONE THING

The first controlled pair yields mystery for triangle and science for circle.

Explain the changed result using the matching lesson.

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Test recommender features for shortcuts, keeping relevant subject information separate from superficial catalogue styling.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Test recommender features for shortcuts, keeping relevant subject information separate from superficial catalogue styling.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record predictions before concluding which clue the classifier uses.

1. Hold topic fixed
2. Hold logo fixed
3. Observe separately
4. Spot confounding

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Observe separately Record predictions before concluding which clue the classifier uses. Either subject information or logo could explain success on the old combinations. Controlled pairs separate their predictions by changing one aspect at a time. Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Practice 2: explain the deciding evidence

Changed-case practice: The first controlled pair yields mystery for triangle and science for circle. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.

2. We can decide the result without examining the changed input or the procedure.
3. That pair is evidence of unwanted logo sensitivity.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That pair is evidence of unwanted logo sensitivity. The subject description stayed mystery; only the irrelevant logo changed.

Repair a missing building block

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?. Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?. Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for -1,0,10,11 and identify the equality boundary for Short.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

